

Table 1. Spanish-input, English-output TruthfulQA transfer results using English-calibrated controllers. Values are mean $\pm$ standard error across five repetitions; TBD marks unrun model-method pairs.

	Method	True (%) $\uparrow$	Informative (%) $\uparrow$	Instruction relevance (0–2) $\uparrow$	Fluency (0–2) $\uparrow$
Gemma-2-2B	Original	51.41 $\pm$ 0.90	86.44 $\pm$ 0.58	1.32 $\pm$ 0.02	1.17 $\pm$ 0.00
	ITI	53.56 $\pm$ 0.47	88.59 $\pm$ 0.45	1.42 $\pm$ 0.01	1.21 $\pm$ 0.01
	ActAdd	63.57 $\pm$ 0.39	67.15 $\pm$ 0.34	0.99 $\pm$ 0.01	0.88 $\pm$ 0.01
	Mean-AcT	48.13 $\pm$ 0.52	90.40 $\pm$ 0.46	1.42 $\pm$ 0.01	1.23 $\pm$ 0.01
	Linear-AcT	47.83 $\pm$ 0.78	90.60 $\pm$ 0.78	1.44 $\pm$ 0.01	1.23 $\pm$ 0.01
	PID-AcT	48.79 $\pm$ 0.61	90.89 $\pm$ 0.49	1.43 $\pm$ 0.01	1.22 $\pm$ 0.01
	ODESteer	57.01 $\pm$ 0.20	86.27 $\pm$ 0.28	1.32 $\pm$ 0.01	1.14 $\pm$ 0.01
	S-PID	55.15 $\pm$ 0.37	86.41 $\pm$ 0.34	1.29 $\pm$ 0.01	1.12 $\pm$ 0.00
	A-LQR	73.81 $\pm$ 0.71	76.55 $\pm$ 0.65	1.12 $\pm$ 0.01	1.06 $\pm$ 0.00
	H_∞ (ours)	77.45 $\pm$ 0.52	64.16 $\pm$ 0.85	1.04 $\pm$ 0.01	1.02 $\pm$ 0.01
Llama-3-8B	Original	42.86 $\pm$ 0.49	94.52 $\pm$ 0.24	1.59 $\pm$ 0.01	1.39 $\pm$ 0.01
	ITI	55.47 $\pm$ 0.88	94.37 $\pm$ 0.20	1.61 $\pm$ 0.02	1.39 $\pm$ 0.01
	ActAdd	43.94 $\pm$ 0.53	93.78 $\pm$ 0.29	1.57 $\pm$ 0.01	1.35 $\pm$ 0.01
	Mean-AcT	46.61 $\pm$ 0.69	94.83 $\pm$ 0.49	1.61 $\pm$ 0.01	1.41 $\pm$ 0.01
	Linear-AcT	47.15 $\pm$ 0.41	94.96 $\pm$ 0.39	1.61 $\pm$ 0.01	1.42 $\pm$ 0.02
	PID-AcT	47.12 $\pm$ 0.63	95.08 $\pm$ 0.51	1.62 $\pm$ 0.01	1.41 $\pm$ 0.01
	ODESteer	95.67 $\pm$ 0.15	11.21 $\pm$ 0.45	0.00 $\pm$ 0.00	0.02 $\pm$ 0.00
	S-PID	52.09 $\pm$ 0.56	94.37 $\pm$ 0.22	1.59 $\pm$ 0.01	1.33 $\pm$ 0.01
	A-LQR	55.57 $\pm$ 0.71	95.03 $\pm$ 0.25	1.56 $\pm$ 0.01	1.27 $\pm$ 0.01
	H_∞ (ours)	85.21 $\pm$ 0.40	64.70 $\pm$ 0.48	0.97 $\pm$ 0.03	0.97 $\pm$ 0.02
Qwen-2.5-14B	Original	TBD	TBD	TBD	TBD
	ITI	TBD	TBD	TBD	TBD
	ActAdd	TBD	TBD	TBD	TBD
	Mean-AcT	TBD	TBD	TBD	TBD
	Linear-AcT	TBD	TBD	TBD	TBD
	PID-AcT	TBD	TBD	TBD	TBD
	ODESteer	TBD	TBD	TBD	TBD
	S-PID	TBD	TBD	TBD	TBD
	A-LQR	TBD	TBD	TBD	TBD
	H_∞ (ours)	TBD	TBD	TBD	TBD
Qwen-2.5-32B	Original	TBD	TBD	TBD	TBD
	ITI	TBD	TBD	TBD	TBD
	ActAdd	TBD	TBD	TBD	TBD
	Mean-AcT	TBD	TBD	TBD	TBD
	Linear-AcT	TBD	TBD	TBD	TBD
	PID-AcT	TBD	TBD	TBD	TBD
	ODESteer	TBD	TBD	TBD	TBD
	S-PID	TBD	TBD	TBD	TBD
	A-LQR	TBD	TBD	TBD	TBD
	H_∞ (ours)	TBD	TBD	TBD	TBD